Active Directory Lab Environment Deployment Report

Platform: Oracle VM VirtualBox

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Project Type: Independent Infrastructure Build

#100DaysOfCybersecurity Project 2

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Introduction

This report documents the end-to-end design, deployment, configuration, troubleshooting, and validation of a functional Active Directory (AD) lab environment built using Oracle VM VirtualBox.

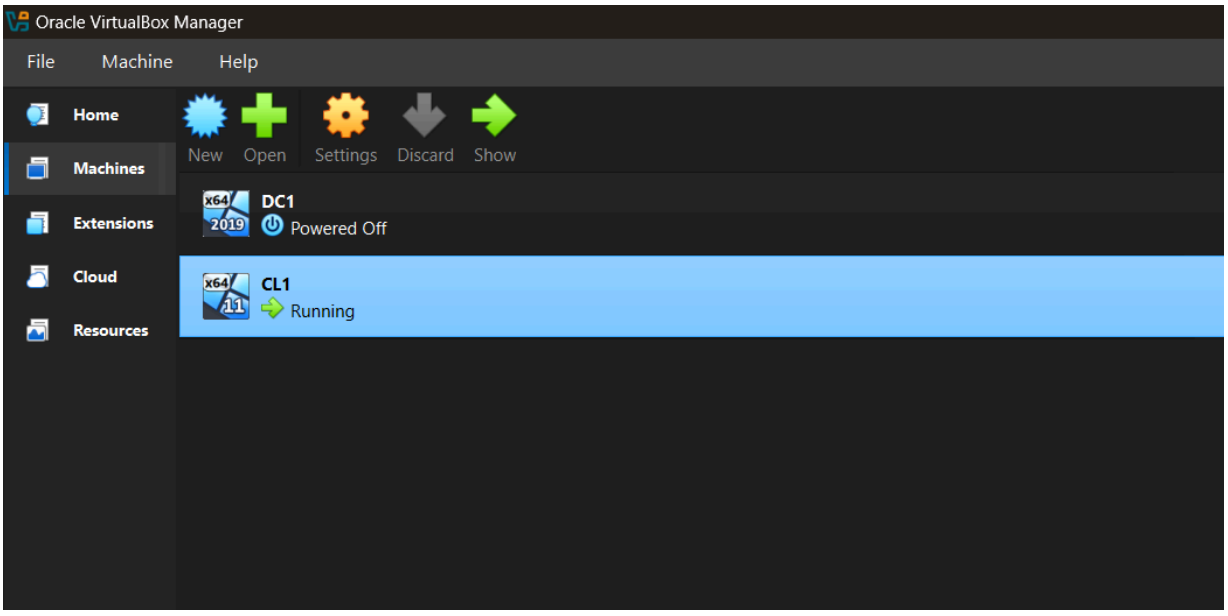
This project was not executed against a fixed lab brief. Instead, it was intentionally designed as a self-directed infrastructure build to reinforce how Active Directory integrates with core network services such as DNS, DHCP, and routing.

This lab emphasized **understanding dependencies**, diagnosing failures, and validating behavior in a controlled environment that closely mirrors real-world enterprise setups, not speed or checklist completion.

Environment Overview

The lab environment was deployed on a single physical host using Oracle VM VirtualBox as the virtualization platform. Two primary virtual machines were created:

- Windows Server 2019 Standard (Desktop Experience)
    - Role: Domain Controller and infrastructure services host
- Windows 11 Pro
    - Role: Domain-joined client workstation



Network Architecture and Design

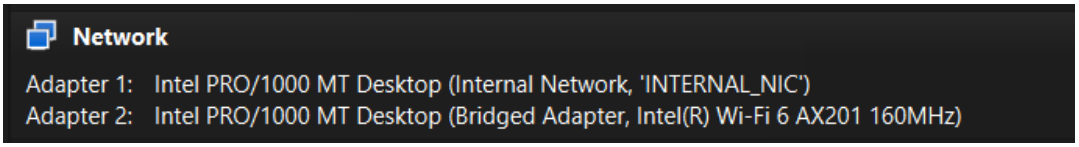
Network Design Rationale

The environment was deliberately designed with network separation, a common enterprise practice.

Server NIC Configuration

The Windows Server VM was configured with two network adapters:

- INTERNAL_NIC:
    - Attached to a VirtualBox Internal Network
    - Used exclusively for domain communication (AD, DNS, DHCP)
    - Assigned a static IP address
- EXTERNAL_NIC:
    - Attached to a bridged adapter
    - Used to provide internet access to the server and routed access for clients

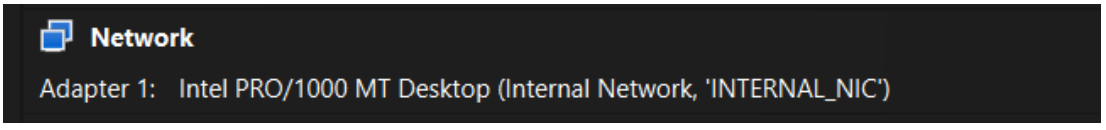


Client NIC Configuration

The Windows 11 client VM was configured with a single NIC, attached to the same Internal Network as the server's internal NIC.

This ensured:

- All authentication traffic stayed internal
- Internet access was only possible through the domain controller
- DNS resolution was centralized



IP Addressing and Initial Connectivity

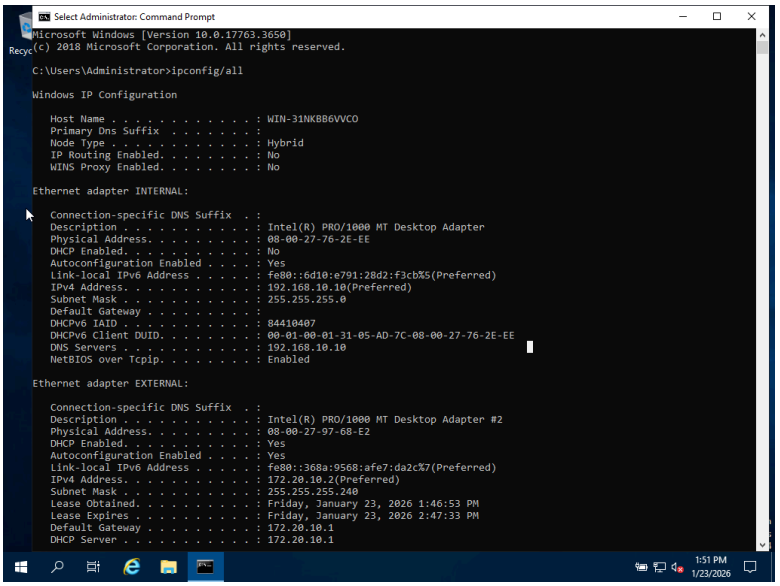
The server's internal NIC was manually assigned a static IP address to ensure stability for directory services.

Config

- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0

- DNS Server: 192.168.10.10

Connectivity and configuration were verified using PowerShell: ipconfig/all.

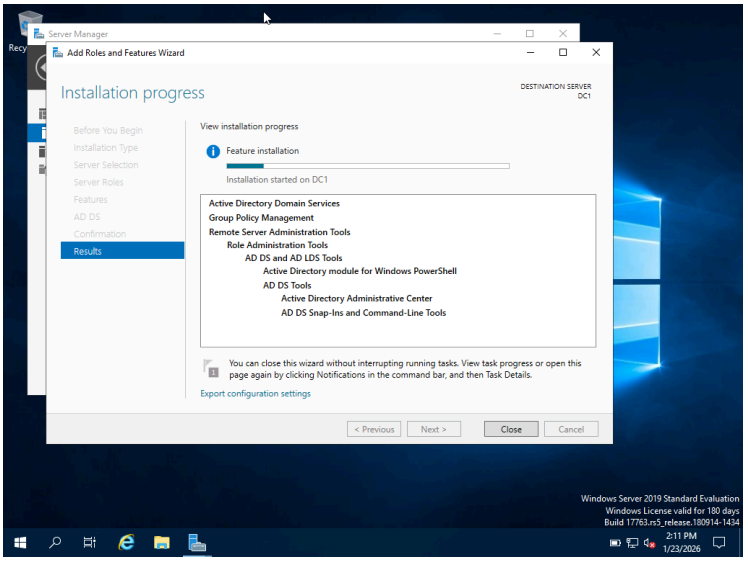


Active Directory Domain Services Deployment

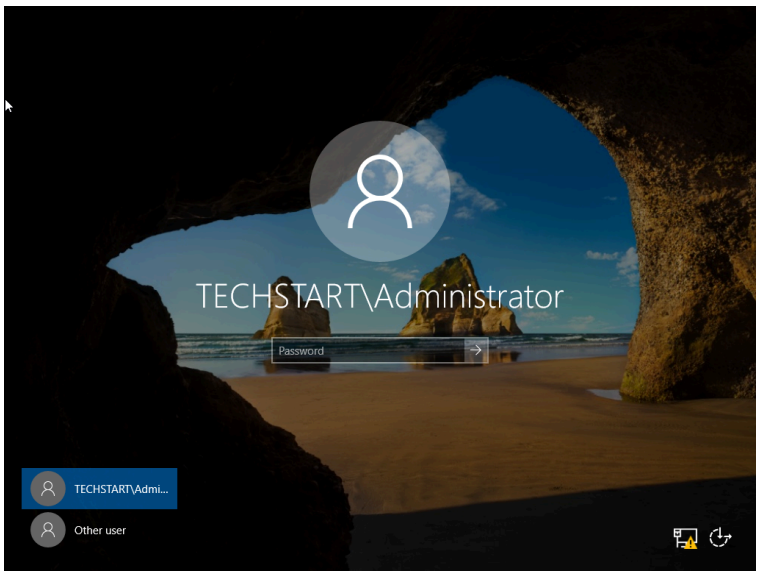
Active Directory Domain Services (AD DS) was installed on the Windows Server 2019 VM using Server Manager.

Following installation:

- The server was promoted to a domain controller
- A new forest and domain were created
- DNS was installed automatically as part of the promotion process



After the promotion, the server rebooted and assumed its role as the authoritative directory and authentication service.



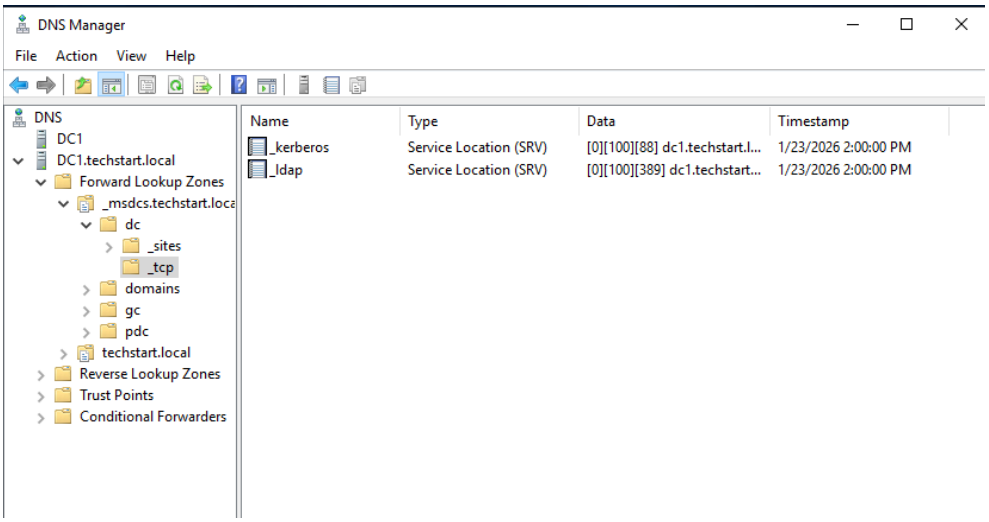
DNS Configuration and Validation

DNS was a critical dependency throughout this lab, as domain controllers rely on name resolution to locate services and authenticate clients.

Post-installation checks confirmed:

- Forward lookup zones were created
- Required SRV records for LDAP and Kerberos were present
- DNS service was running and listening on the INTERNAL_NIC.

Early domain-join attempts failed, producing SRV lookup errors. These errors reinforced the principle that Active Directory failures often originate from DNS or network misconfiguration, not the directory service itself.



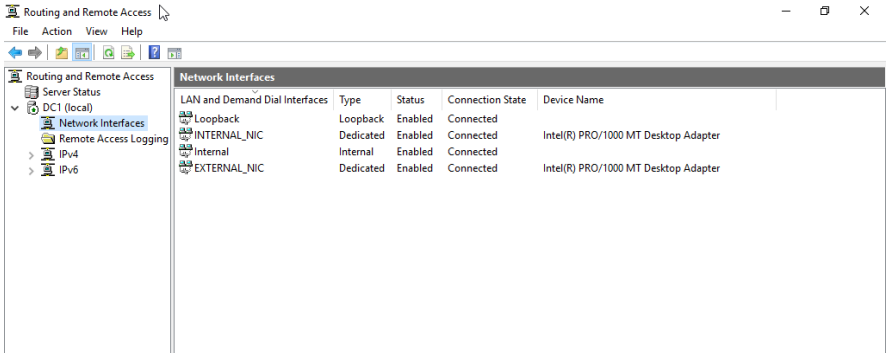
Routing and Remote Access Services (RRAS) - NAT Configuration

Routing and Remote Access Services (RRAS) was installed to allow internal clients to access the Internet.

Configuration steps included:

- Enabling RRAS via the setup wizard
- Selecting NAT routing
- Assigning:
    - External NIC as the internet-facing interface
    - Internal NIC as the private interface (handling domain traffic)

This allowed internal domain clients to share the server's external connection.



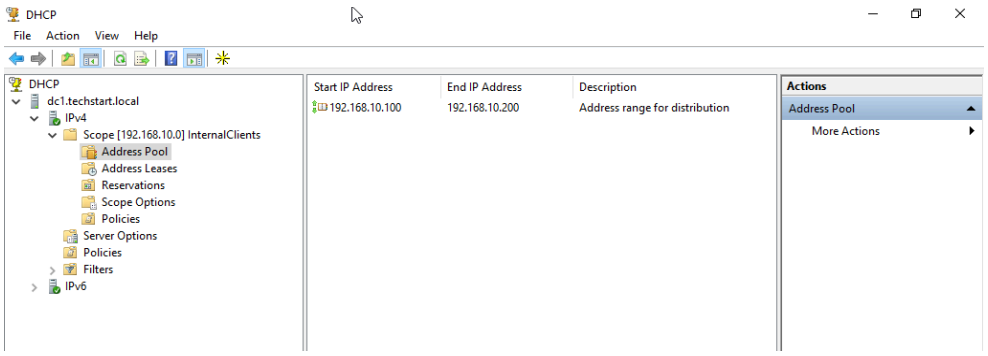
DHCP Configuration

The DHCP role was installed to automate IP address assignment for domain clients.

A DHCP scope was created, defining:

- IP address range and Subnet mask
- Default gateway (domain controller)
- DNS server (domain controller)

This eliminated manual client configuration and ensured consistent DNS behavior across domain devices.



Active Directory Administration

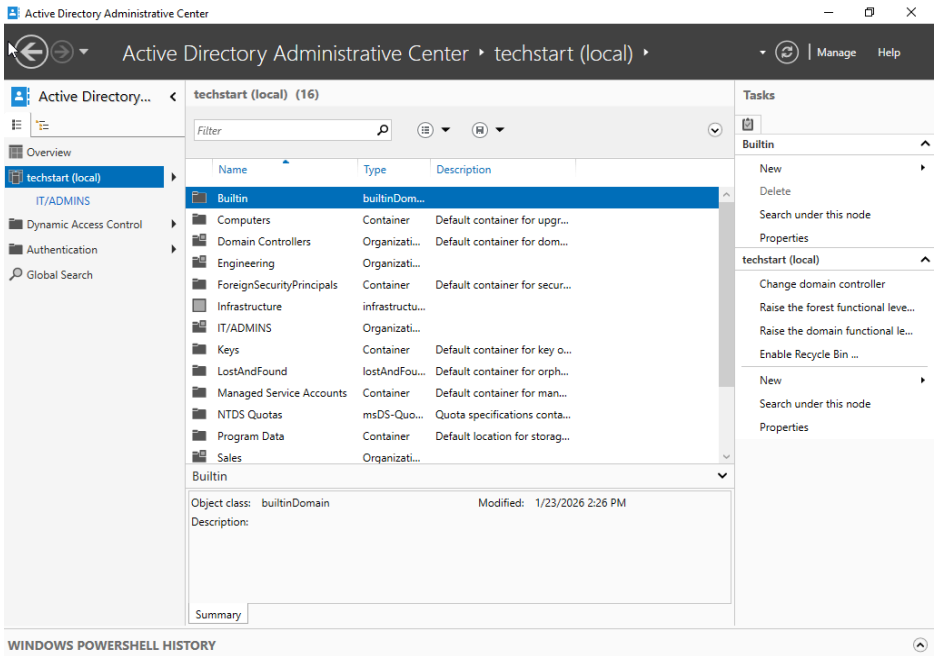
Using the Active Directory Administrative Center (ADAC), the following were created:

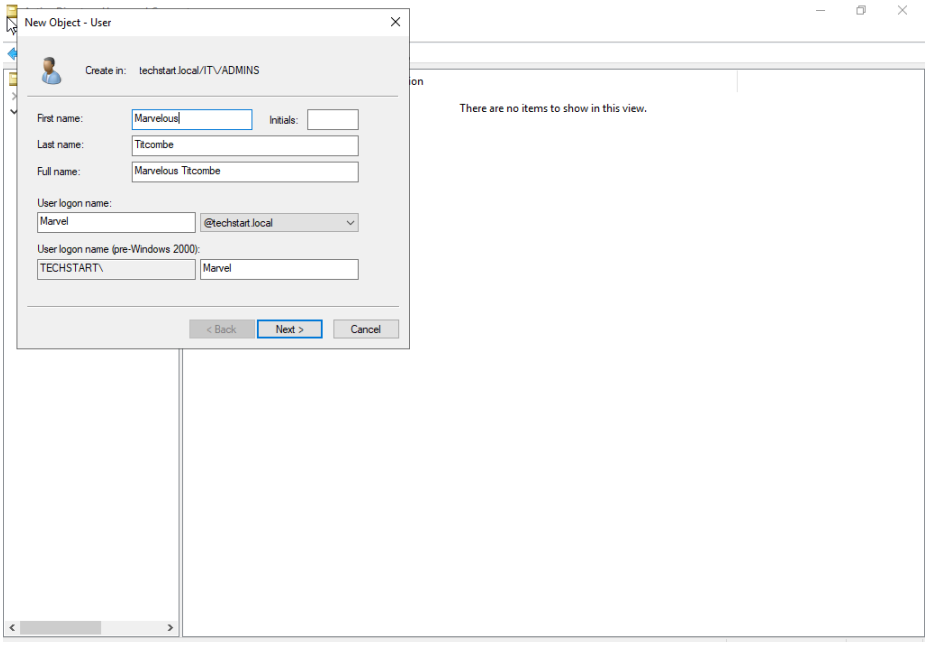
- Organizational Units (OUs)
- User accounts for authentication testing

Accounts were initially configured with:

- "User must change password at next logon."

This setting later surfaced as a domain-join and login constraint, requiring adjustment during troubleshooting.





Client Domain Join - Issues and Resolution

Initial Failures

Several issues prevented the client from joining the domain:

- Incorrect Windows edition (non-Pro)
- DNS SRV record lookup failures
- NIC attachment mismatch between client and server
- Password policy preventing authentication

Root Cause: NIC Misalignment

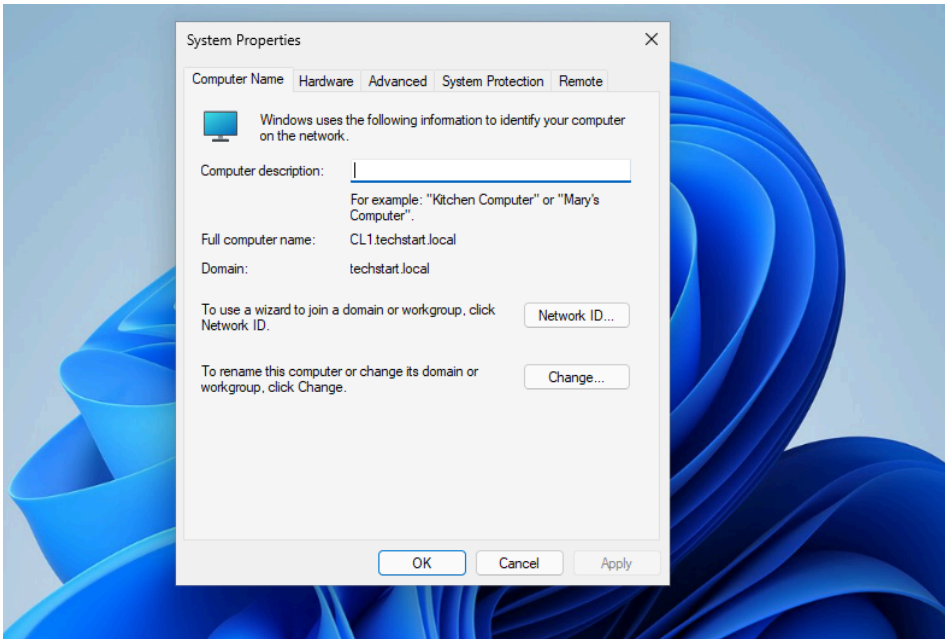
Although both machines had internet access, the client was initially attached to the wrong VirtualBox network. Once the client's NIC was correctly attached to the internal network, DNS resolution and domain discovery succeeded.

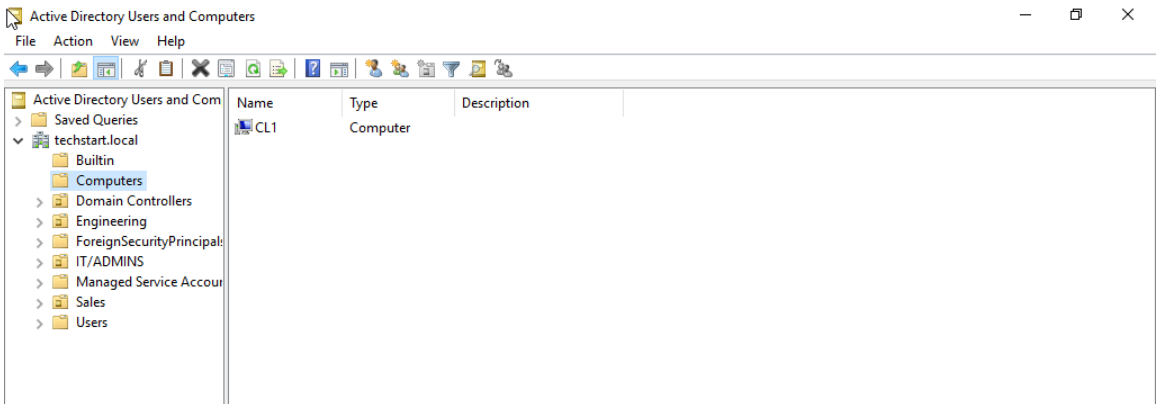
Password Policy Resolution

The account used for joining the domain had “must change password at next logon” enabled. This prevented successful authentication. The issue was resolved by:

- Logging in once with an administrative context
- Changing the password
- Re-attempting the domain join

After resolution, the client successfully joined the domain and appeared in Active Directory Users and Computers.

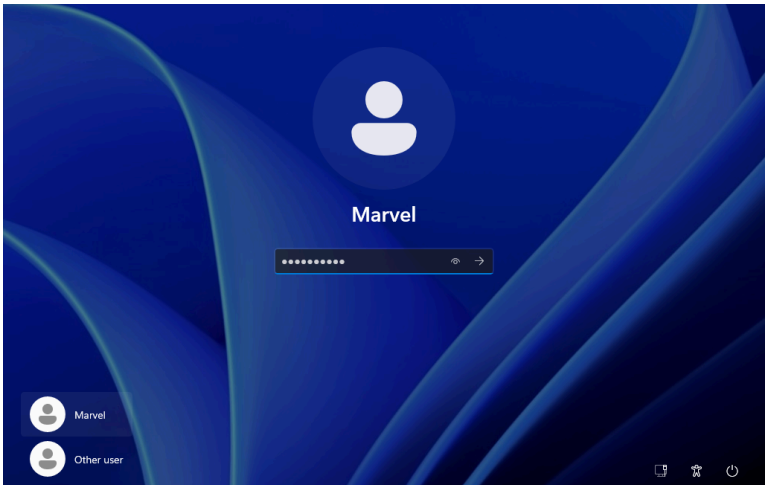


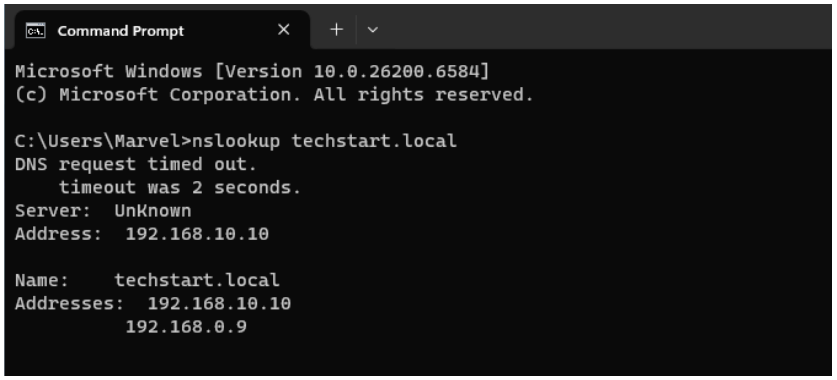


Validation and Testing

Validation included:

- Successful domain user login
- DNS name resolution
- Internet access from the domain client
- Client object visibility in AD





Lessons Learned

- Active Directory is inseparable from DNS correctness.
- NIC attachment and network segmentation matter more than expected.
- Password policies can block workflows if not considered.
- Most AD issues are network problems in disguise.

Challenges Encountered and Resolved

This lab took three days due to:

- OS edition limitations
    - This was resolved by reinstalling the client using Windows 11 Pro.
- DNS resolution failures
- NIC misconfiguration
    - Adding and configuring a second adapter resolved routing and connectivity issues.
- Authentication constraints

Each issue reinforced real-world troubleshooting discipline.

Conclusion

This project delivered a fully functional Active Directory environment built without a prescriptive guide. By designing, breaking, fixing, and validating the environment, this lab strengthened both technical understanding and systems-level thinking.

The environment now serves as a reusable platform for future experimentation in security, policy, and identity management.